

DSA210 PROJECT PROPOSAL

In this project, I aim to analyze ranked gameplay data from *League of Legends* to investigate the factors that influence match outcomes. Specifically, I will examine how performance metrics such as KDA, damage, vision score, and player tier are associated with winning or losing a match.

The dataset will be collected using the Riot Games API (<https://developer.riotgames.com/apis>). Instead of relying on a single player, I will gather data from leaderboard players (Challenger, Grandmaster, and Master tiers) to ensure a sufficiently large and diverse dataset. Each observation will represent a player's performance in a single ranked match. I expect to collect several thousand observations, which is sufficient for meaningful statistical analysis and hypothesis testing.

The dataset will include variables such as match outcome (win/loss), kills, deaths, assists (KDA), champion played, role, damage dealt, vision score, and player tier. Additional features such as per-minute statistics will be derived during preprocessing to enrich the analysis.

Exploratory Data Analysis (EDA) will be conducted to understand the distribution and relationships between variables using summary statistics and visualizations. I will also perform hypothesis testing to evaluate specific research questions. For example:

- **H0:** The mean vision score is the same for winning and losing matches.
- **H1:** The mean vision score is higher in winning matches.
- **H0:** Match outcome is independent of player tier.
- **H1:** Match outcome depends on player tier.

Finally, I will apply basic machine learning methods (e.g., logistic regression) to explore whether match outcomes can be predicted from performance metrics.

This project follows a complete data science pipeline, including data collection, preprocessing, exploratory analysis, hypothesis testing, and predictive modeling.